

Dietary lipids and blood cholesterol: quantitative meta-analysis of metabolic ward studies.



OBJECTIVE: To determine the quantitative importance of dietary fatty acids and dietary cholesterol to blood concentrations of total, low density lipoprotein, and high density lipoprotein cholesterol. DESIGN: Meta-analysis of metabolic ward studies of solid food diets in healthy volunteers. SUBJECTS: 395 dietary experiments (median duration 1 month) among 129 groups of individuals. RESULTS: Isocaloric replacement of saturated fats by complex carbohydrates for 10% of dietary calories resulted in blood total cholesterol falling by 0.52 (SE 0.03) mmol/l and low density lipoprotein cholesterol falling by 0.36 (0.05) mmol/l. Isocaloric replacement of complex carbohydrates by polyunsaturated fats for 5% of dietary calories resulted in total cholesterol falling by a further 0.13 (0.02) mmol/l and low density lipoprotein cholesterol falling by 0.11 (0.02) mmol/l. Similar replacement of carbohydrates by monounsaturated fats produced no significant effect on total or low density lipoprotein cholesterol. Avoiding 200 mg/day dietary cholesterol further decreased blood total cholesterol by 0.13 (0.02) mmol/l and low density lipoprotein cholesterol by 0.10 (0.02) mmol/l. CONCLUSIONS: In typical British diets replacing 60% of saturated fats by other fats and avoiding 60% of dietary cholesterol would reduce blood total cholesterol by about 0.8 mmol/l (that is, by 10-15%), with four fifths of this reduction being in low density lipoprotein cholesterol.